

**TCS3351 CRYPTOGRAPHY AND DATA SECURITY**  
**Tutorial 10 - Cryptographic Hash Functions/Message Authentication**  
**Codes**

1. Briefly explain how the hash function can be used for message authentication without the usage of encryption function.
2. Briefly explain the man-in-the middle attack on the hash function and how it can be prevented?
3. State the difference between weak and strong collision resistance in the case of cryptographic hash functions.
4. Briefly explain the model in which the public-key encryption can be used for confidentiality, authentication, and digital signature
5. Briefly explain how the message authentication and confidentiality can be provided by Message Authentication Code (MAC) with authentication tied to plaintext and how it is different from authentication tied to cipher text.

**References:**

1. William Stallings, “Cryptography and Network Security – Principles and Practice”, Seventh Edition, Prentice Hall, 2017
2. Forouzan, B. A. & Mukhopadhyay, D. (2015). Cryptography & Network Security (3rd ed.). McGraw-Hill.